

Pattern Classification In Recognizing Qalqalah Kubra Pronunciation Using Multilayer Perceptrons

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Abstract—Quranic recitations require precise pronunciation in its recitation. Because of this, Tajweed is important as a set of rules that govern how certain verses must be pronounced. One of the many Tajweed rules is called Qalqalah. The voice signals of Qalqalah Kubro (QK) (one of the Qalqalah variations) pronunciation have distinct patterns which can be recognized with pattern classification algorithms such as Multilayer Perceptron (MLP). This study investigates the performance of the MLP in identifying correct pronunciation of QK of a reader. The pronunciation sound waves of QK were first divided into equal length segments. Next, important features were extracted using Mel Frequency Cepstrum Coefficient (MFCC) analysis. After training, the MLP performance was analyzed to discriminate between correct and incorrect pronunciations. Results show that the MLP classifier trained using the MFCC features was able to accurately distinguish between the two cases.

Keywords—Qalqalah Kubro; Multilayer Perceptron; Mel Frequency Cepstrum Coefficient

I. INTRODUCTION

Speech is an important part in human communication. It varies widely in terms of accent, pronunciation, articulation, roughness, nasality, pitch, volume and speech. Furthermore, speech patterns can be further distorted by external environmental features such as background noise and echoes.

Tajweed is an Arabic word meaning correct pronunciation during recitation. It is a set of rules which dictate how the Qur'an should be read. Tajweed is derived from the root word 'ja-wa-da', meaning to make well, make better or improve. It is important to read the Qur'an with tajweed as the wrong pronunciation may change the meaning completely [1].

Qalqalah is one of the many Tajweed rules. Qalqalah requires that a certain phrase be read with a slight echo or bouncing sound. This rule is further divided into two: Qalqalah Sughro and Qalqalah Kubro (QK).

This paper describes the use of Multi-Layer Perceptron (MLP) neural network to classify correct pronunciation of

Qalqalah Kubro using features extracted from Mel Frequency Cepstral Coefficient (MFCC) analysis.

MFCC is a well-established speech features extraction technique for many speech recognition systems [2]. The strength of MFCC is its cepstral features, which are relatively more compact and discriminative, compared to other representation methods. Therefore, they are able to provide higher baseline performance over filter bank features [3].

The organization of this paper is as follows: the theoretical background is presented in Section II. Section III focuses on the methodology used, while Section IV presents the results and discussions. Finally, concluding remarks are presented in Section V.

II. THEORY

A. Mel-Frequency Cepstrum (MFC) Analysis

MFC analysis is an algorithm to extract features from audio signals. It takes human perception sensitivity with respect to frequencies into consideration and thus is most appropriate for voice recognition [4, 5]. The coefficients comprise a good representation of the dominant features in acoustic information for selected windows of time.

For a signal with N frames, $n = 0, 1, 2, \dots, N - 1$, the MFC coefficient is defined as:

$$c(n) = DCT(\log(|FFT(s(n))|)) \quad (1)$$

where, $c(n)$ is the MFC coefficients at frame n ; $s(n)$ is the original signal at frame n , after application of pre-filtering and some windowing method.

Six steps are involved to extract features using the MFCC technique, as shown in Figure 1. The first step is Pre-Emphasis Filter to spectrally flatten the signal. Then the pre-emphasized speech signal is separated into short segments called frames. The second step to decompose the input signal into overlapping time windows. This is done to comprehensively capture the signal's temporal features and changes. As a result of frame blocking, high

frequency components are produced at end of every signal block, which is known as *leakage effect* in the spectrum. To minimize this effect, the Hamming window is then used to maintain the continuity of the first and last points in the frame.

After windowing, Fourier analysis is performed on each frame, resulting in short-time Discrete Fourier Transform (DFT). The spectrum of each frame is filtered by a set of filters, and the power of each band is calculated. To obtain a good frequency resolution, a 256-point FFT is used [6]. The filter bank consists of 26 triangular shaped band pass filter. The mapping from linear frequency to Mel-Frequency is shown in Figure 2 using Equation 2.

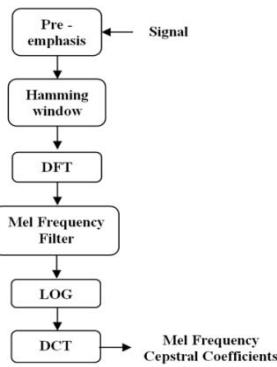


Figure 1 Workflow to generate MFCC features from input signal.

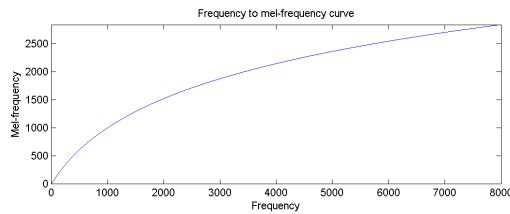


Figure 2 Relationship between mel and linear frequencies

$$mel = 2595(\log_{10}(1 + freq/700)) \quad (2)$$

The mel-frequency cepstral coefficients are then derived by taking the logarithm of the band-passed frequency response and calculating the Discrete Cosine Transform (DCT) for each intermediate signal.

III. METHODOLOGY

A. Sample Description

The 100 samples for QK pronunciation was obtained from same voice record with five different types of pronunciation. The dataset include 50 samples each with and without correct pronunciation. The signals were pre-processed with pre-emphasis filter to increase the amplitude of the signals.

B. MFCC Feature Extraction

MFCC analysis was used to extract features in the signal. Segmented signals were first divided into equal-

length frames. The sampling frequency used was 8 kHz. In order to smoothen the frame-to-frame transition, 50% of each successive frame was overlapped. Each block was then multiplied with a Hamming window in order to preserve the continuity between the first and last points of each frame.

Then, Fast Fourier Transform (FFT) was used to convert the signal into frequency domain. Mel-scale was applied to this transformed signal using 26 rectangular filter banks.

Next, the logarithm of the amplitude spectrum within the frequency scale was computed. The filter banks were then used to convert the frequency scale to mel-frequency scale. The dimension of the MFC feature coefficients was 12. Finally, Discrete Cosine Transform (DCT) was applied to reduce the number of parameters in the derived signal and to produce a set of MFC coefficients.

A three-layer MLP structure was used for classification, with sigmoid transfer functions in both the hidden and output layers. The hidden units used were 5. The scaled conjugate gradient was used as the training algorithm.

Classification Accuracy (CA) was then calculated using Eq. (3). CA is the percentage of correctly classified cases divided by number of test cases:

$$CA_i = \frac{TP + TN}{N} \times 100 \quad (3)$$

where:

TP = Number of true positives

TN = Number of true negatives

N = Number of test cases

IV. RESULT & DISCUSSION

A. Result of MFCC Extraction Process

Figure 3 to Figure 7 shows the MFCC feature extraction results for arabic phrases containing the five QK letters.

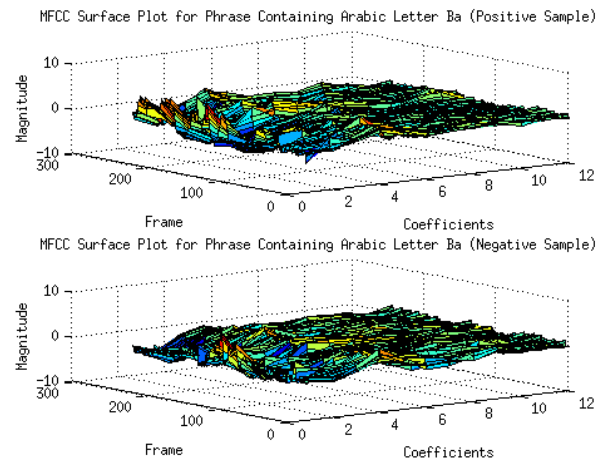


Figure 3 MFCC surface plot for phrase containing the letter ب.

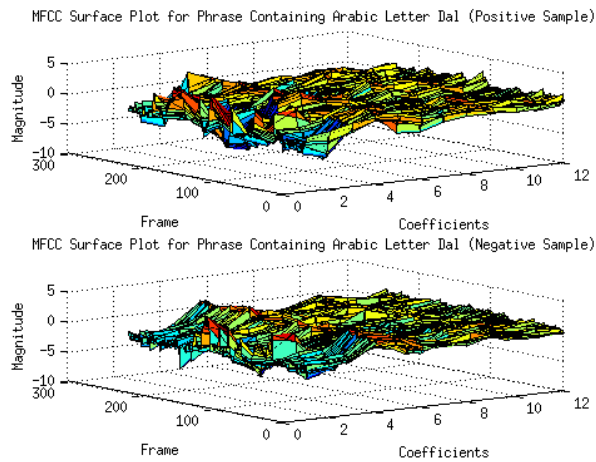


Figure 4 MFCC surface plot for phrase containing the letter د.

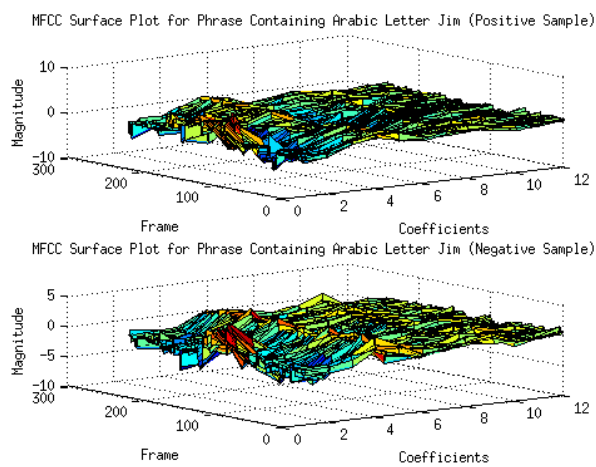


Figure 5 MFCC surface plot for phrase containing the letter ج.

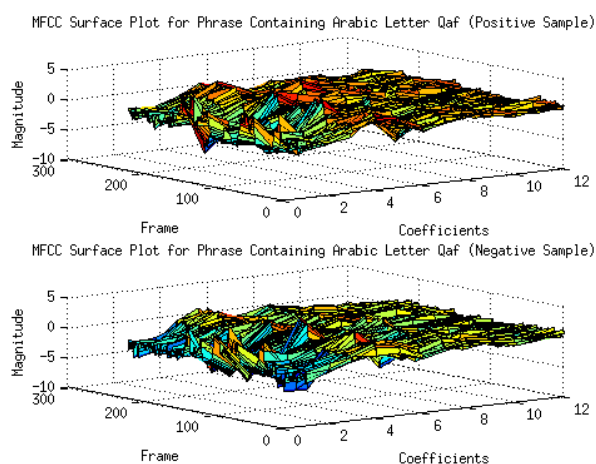


Figure 6 MFCC surface plot for phrase containing the letter ق

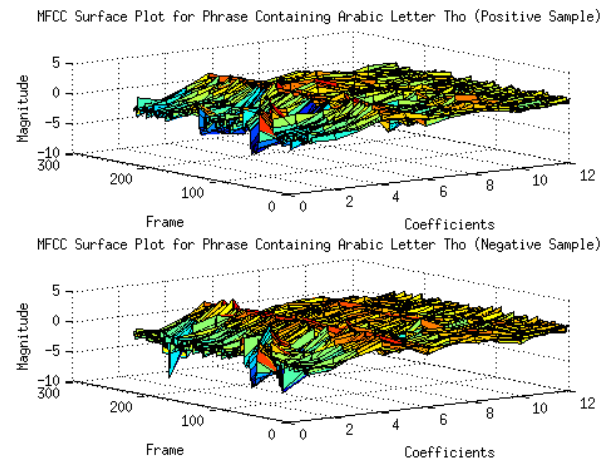


Figure 7 MFCC surface plot for phrase containing the letter ط

B. Result of Classification Process

Figure 8 to Figure 12 shows the result for five pronunciations of QK. These 20 testing samples were collected from 100 samples consisting of correct and incorrect pronunciations.

The output and target is compared using threshold acquired from Receiver Operating Characteristic (ROC) analysis. Any output below the threshold value was set to incorrect pronunciation, while outputs above the threshold value were set to correct pronunciation. If the output and target value are separate between thresholds, it mean the sample data is wrongly classified.

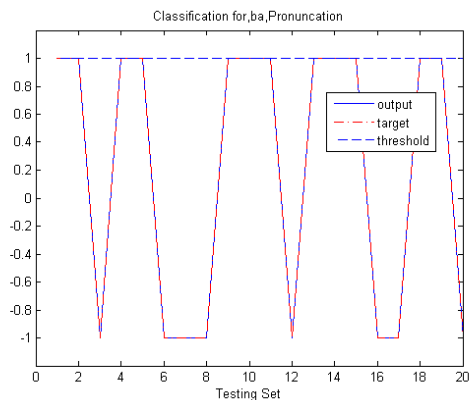


Figure 8 Classification result for test case إِذَا وَقَب

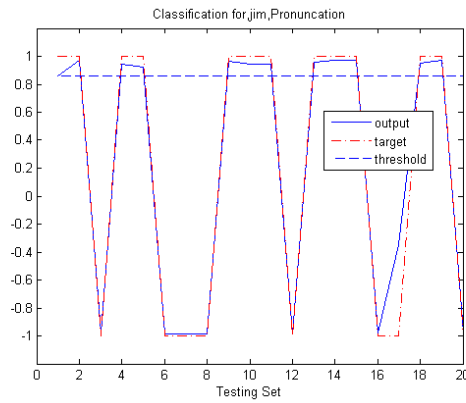


Figure 9 Classification result for test case الجُور

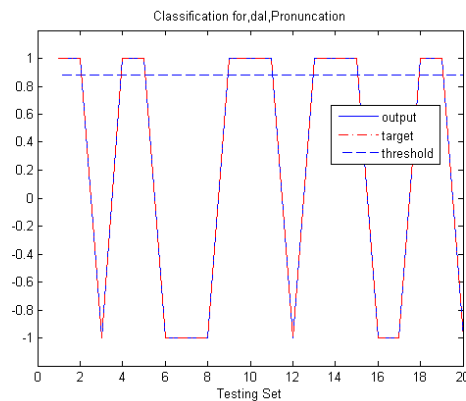


Figure 10 Classification result for test case إِذَا حَسَدَ

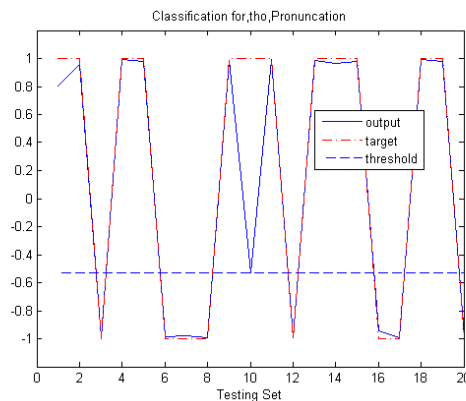


Figure 11 Classification result for test case مُحِيطَ

Table 1 shows the classification accuracy and threshold for each result. It can be seen that 3 of 5 test cases achieved 100% classification accuracy. The other two cases also achieved high accuracy at 95%. Furthermore, the threshold values were close to 1, indicating good discriminative ability for the classifier.

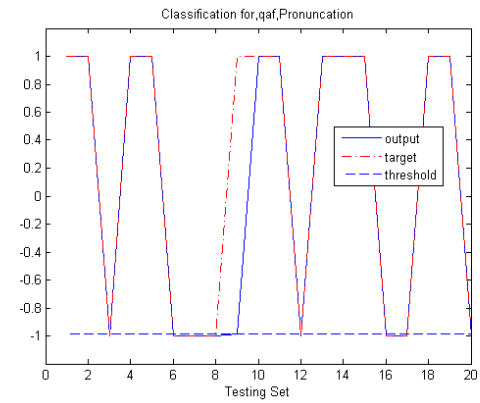


Figure 12 Classification result for test case مَا خَلَقَ

TABLE 1
CLASSIFICATION ACCURACY AND THRESHOLD FOR QK PRONUNCIATION

CA (%)	Threshold	QK
100	0.9994	إِذَا وَقَبَ
100	0.8593	الْجُورَ
100	0.8836	إِذَا حَسَدَ
95	-0.5270	مُحِيطَ
95	-0.9847	مَا خَلَقَ

V. CONCLUSION

The use of the MLP in classifying QK has been discussed in this paper. Based on the results obtained from this study, MLP trained with MFCC features had been proven to be able to differentiate between correct and incorrectly Qalqalah Kubra pronounced.

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